

Supplementary material

Table 1: Supplementary Table S1. Objective pre-model exclusion statistics for tests not used in resistance-HI label training.

Test	$R_{ds,ref}$	$\Delta R_{ds,95}$	Trend	Reason
1	1.3373	0.00354	-0.098	insufficient observable Rds degradation
2	116.5270	0.15992	-0.037	invalid on-state Rds reference
16	1.4890	0.00414	-0.333	insufficient observable Rds degradation
17	1.5108	0.00004	-0.703	insufficient observable Rds degradation
22	1.3983	0.00957	0.605	insufficient observable Rds degradation
27	1.8409	-0.00008	-0.324	insufficient observable Rds degradation
31	1.6078	-0.00002	-0.185	insufficient observable Rds degradation
41	1.8197	-0.00007	-0.200	insufficient observable Rds degradation

Table 2: Supplementary Table S2. PG-RGRU random-seed repeat using the fixed 64-hidden/90-epoch setting.

Seed	HI MAE	HI RMSE	Warnings	EOL MAE (h)	RUL MAE (h)
1	0.0206	0.0362	7/7	0.1713	0.1190
7	0.0211	0.0328	7/7	0.2276	0.1407
21	0.0160	0.0333	7/7	0.1668	0.1154
42	0.0216	0.0346	7/7	0.0563	0.0491
100	0.0210	0.0325	7/7	0.2280	0.1399

Table 3: Supplementary Table S3. Summary statistics of the five-seed PG-RGRU repeat.

Metric	Mean	SD	Min	Max
HI MAE	0.0201	0.0023	0.0160	0.0216
HI RMSE	0.0339	0.0015	0.0325	0.0362
Warning rate	1.0000	0.0000	1.0000	1.0000
EOL MAE (h)	0.1700	0.0700	0.0563	0.2280
RUL MAE (h)	0.1128	0.0374	0.0491	0.1407

Table 4: Supplementary Table S4. Simulink–Python consistency for the exported 64-hidden PG-RGRU checkpoint.

Test	HI MAE	HI RMSE	Max diff.	EOL diff. (h)	Hit
6	0.0226	0.0304	0.0440	0.0000	Yes
13	0.0036	0.0101	0.0682	0.0000	Yes
14	0.0033	0.0101	0.0768	0.0000	Yes
20	0.0010	0.0054	0.0621	0.0000	Yes
25	< 0.0001	0.0005	0.0132	0.0000	Yes
34	< 0.0001	< 0.0001	< 0.0001	0.0000	Yes
42	< 0.0001	< 0.0001	< 0.0001	0.0000	Yes

Table 5: Supplementary Table S5. Three-seed 64-hidden warning-probability decision summary.

Metric	Raw mean	Raw SD	Calibrated mean	Calibrated SD
Warning coverage	7.000	0.000	7.000	0.000
Missed warning tests	0.000	0.000	0.000	0.000
False-before-target tests	5.667	0.577	2.000	0.000
False-warning windows	0.1103	0.0019	0.0629	0.0209
Lead-time error MAE (h)	1.2875	0.0222	1.2263	0.0012

Table 6: Supplementary Table S6. Threshold-derived EOL relative-error intervals on the seven test trajectories.

EOL relative error interval	Tests	Proportion
0–10%	7	100.0%
10–20%	0	0.0%
20–30%	0	0.0%
>30%	0	0.0%

Table 7: Supplementary Table S7. Test-level threshold-derived EOL relative error.

Test	True EOL (h)	Absolute error (h)	Relative error (%)
6	0.7210	0.0575	7.98
13	1.7439	0.0341	1.95
14	20.1151	0.1068	0.53
20	8.0377	0.1046	1.30
25	11.1711	0.0420	0.38
34	41.8143	0.0348	0.08
42	0.5485	0.0144	2.62